

Mathematical Details of the GLW Joint Interval-DMR Model

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1 Mathematical Details of the Joint Interval-DMR Model

1.1 Observed Data and Indexing

Let $i = 1, \dots, N$ index patients. Patient i has n_i longitudinal molecular observations and K_i at-risk intervals for first deep molecular response (DMR). The cleaned GLW data contain $N = 87$ patients, $\sum_i n_i = 495$ longitudinal observations, and $\sum_i K_i = 275$ at-risk intervals.

For longitudinal visit j of patient i , define

$$y_{ij} = \log\text{-MRD}_{ij}, \quad t_{ij} = \text{years from imatinib start}, \quad x_{ij} = \log(1 + t_{ij}),$$

and let $z_{ij} = 1$ if the sample type is bone marrow and $z_{ij} = 0$ otherwise. The transformed time x_{ij} is used to allow rapid early molecular change and slower later change.

For at-risk interval k of patient i , define

$$L_{ik} = \text{interval start}, \quad R_{ik} = \text{interval end}, \quad \Delta_{ik} = R_{ik} - L_{ik},$$

where time is measured in years. Let

$$t_{ik}^{mid} = \frac{L_{ik} + R_{ik}}{2}, \quad x_{ik}^{mid} = \log(1 + t_{ik}^{mid}).$$

The event indicator is

$$d_{ik} = \begin{cases} 1, & \text{if first observed DMR occurs in interval } (L_{ik}, R_{ik}], \\ 0, & \text{otherwise.} \end{cases}$$

By construction, each patient has at most one interval with $d_{ik} = 1$. Intervals before first DMR or through censoring have $d_{ik} = 0$.

1.2 Longitudinal Molecular Submodel

The latent patient-specific molecular trajectory is

$$\mu_{ij} = \beta_0 + \beta_1 x_{ij} + \beta_2 x_{ij}^2 + \beta_3 z_{ij} + b_{0i} + b_{1i} x_{ij}.$$

Here β_0 is the population intercept, β_1 and β_2 describe the population time trend, and β_3 adjusts for sample type. The patient-specific random intercept and random slope are

$$\mathbf{b}_i = \begin{pmatrix} b_{0i} \\ b_{1i} \end{pmatrix} \sim N_2(\mathbf{0}, \Sigma_b),$$

with

$$\Sigma_b = \begin{pmatrix} \tau_0 & 0 \\ 0 & \tau_1 \end{pmatrix} \Omega_b \begin{pmatrix} \tau_0 & 0 \\ 0 & \tau_1 \end{pmatrix},$$

where $\tau_0, \tau_1 > 0$ are random-effect standard deviations and Ω_b is a 2×2 correlation matrix.

For observations not reported at the assay floor,

$$y_{ij} \mid \mu_{ij}, \sigma_y \sim N(\mu_{ij}, \sigma_y^2).$$

The Stan model treats values at the reporting floor as left-censored. Let $c = -5$ be the log-MRD floor and let $a_{ij} = 1$ indicate that the observation is at the floor. Then the longitudinal contribution is

$$\ell_{ij}^Y = \begin{cases} \log \phi(y_{ij}; \mu_{ij}, \sigma_y^2), & a_{ij} = 0, \\ \log \Phi\left(\frac{c - \mu_{ij}}{\sigma_y}\right), & a_{ij} = 1, \end{cases}$$

where $\phi(\cdot; \mu, \sigma^2)$ is the normal density and $\Phi(\cdot)$ is the standard normal cumulative distribution function.

1.3 Interval-DMR Submodel

The event submodel links DMR onset to the latent molecular trajectory. At the midpoint of interval k , the latent molecular value used by the hazard model is

$$\mu_i(t_{ik}^{mid}) = \beta_0 + \beta_1 x_{ik}^{mid} + \beta_2 (x_{ik}^{mid})^2 + b_{0i} + b_{1i} x_{ik}^{mid}.$$

The current Stan implementation does not include sample type in the interval midpoint trajectory because intervals are defined at the patient-follow-up level rather than at a specific sample type.

The interval-specific log hazard is

$$\eta_{ik} = \gamma_0 + \gamma_1 x_{ik}^{mid} + \gamma_2 \log(1 + \Delta_{ik}) + \alpha \mu_i(t_{ik}^{mid}),$$

and the interval hazard is

$$h_{ik} = \exp(\eta_{ik}).$$

Assuming a piecewise-constant hazard within each observed interval, the cumulative hazard over the interval is

$$H_{ik} = h_{ik} \Delta_{ik}.$$

Therefore, the probability that first DMR occurs during the interval is

$$p_{ik} = P(d_{ik} = 1 \mid \text{at risk}) = 1 - \exp(-H_{ik}).$$

The interval likelihood contribution is

$$\ell_{ik}^D = d_{ik} \log\{1 - \exp(-H_{ik})\} - (1 - d_{ik})H_{ik}.$$

This is exactly the likelihood form implemented in Stan:

$$\ell_{ik}^D = \begin{cases} \log\{1 - \exp(-H_{ik})\}, & d_{ik} = 1, \\ -H_{ik}, & d_{ik} = 0. \end{cases}$$

1.4 Joint Likelihood

Let θ denote all fixed-effect, variance, association, and random-effect parameters. Conditional on the random effects, the joint likelihood is

$$L(\theta) = \prod_{i=1}^N \left[\prod_{j=1}^{n_i} L_{ij}^Y(\theta) \prod_{k=1}^{K_i} L_{ik}^D(\theta) \right],$$

or equivalently the log-likelihood is

$$\ell(\theta) = \sum_{i=1}^N \left[\sum_{j=1}^{n_i} \ell_{ij}^Y(\theta) + \sum_{k=1}^{K_i} \ell_{ik}^D(\theta) \right].$$

The shared random effects b_{0i} and b_{1i} induce dependence between the repeated molecular measurements and first-DMR process.

1.5 Prior Distributions

The Stan implementation uses weakly informative priors centered on clinically plausible molecular-response behavior:

$$\begin{aligned} \beta_0 &\sim N(-2.5, 2^2), & \beta_1 &\sim N(-1, 1^2), & \beta_2 &\sim N(0, 0.5^2), \\ \beta_3 &\sim N(0, 1^2), & \sigma_y &\sim \text{Exponential}(1). \end{aligned}$$

For the interval-DMR submodel,

$$\begin{aligned} \gamma_0 &\sim N(-2, 2^2), & \gamma_1 &\sim N(0, 1^2), & \gamma_2 &\sim N(0, 1^2), \\ \alpha &\sim N(-0.5, 0.75^2). \end{aligned}$$

The prior mean for α is negative because lower log-MRD corresponds to deeper molecular response; therefore lower molecular burden is expected to increase the probability of DMR within an interval.

For the random effects,

$$\tau_0, \tau_1 \sim \text{Exponential}(1), \quad \Omega_b \sim \text{LKJ}(2).$$

The Stan code uses a non-centered Cholesky parameterization:

$$\mathbf{b}_i = \text{diag}(\boldsymbol{\tau}) L_{\Omega} \mathbf{z}_i, \quad \mathbf{z}_i \sim N_2(\mathbf{0}, I_2),$$

where $L_{\Omega} L_{\Omega}^T = \Omega_b$.

1.6 Posterior Distribution

The posterior distribution is proportional to the product of the joint likelihood and all prior densities:

$$p(\theta \mid Y, D) \propto \left[\prod_{i=1}^N \prod_{j=1}^{n_i} L_{ij}^Y(\theta) \prod_{k=1}^{K_i} L_{ik}^D(\theta) \right] p(\boldsymbol{\beta}) p(\boldsymbol{\gamma}) p(\alpha) p(\sigma_y) p(\boldsymbol{\tau}) p(\Omega_b) \prod_{i=1}^N p(\mathbf{b}_i \mid \boldsymbol{\tau}, \Omega_b).$$

Posterior inference focuses on the molecular time trend (β_1, β_2) , between-patient heterogeneity $(\tau_0, \tau_1, \Omega_b)$, visit-gap effect γ_2 , and association parameter α .

1.7 Interpretation of Key Parameters

- β_1, β_2 : population-level change in log-MRD over time.
- β_3 : systematic difference between bone marrow and non-bone-marrow samples.
- τ_0, τ_1 : between-patient heterogeneity in baseline molecular level and response slope.
- γ_1 : change in interval-DMR hazard across follow-up time.
- γ_2 : association between visit-gap length and interval event probability after accounting for the interval length.
- α : association between latent molecular burden and DMR onset. A negative value supports the clinical expectation that lower log-MRD increases the chance of DMR.

1.8 Derived Quantities

For each interval, the generated quantity

$$\hat{p}_{ik} = 1 - \exp(-H_{ik})$$

summarizes the posterior interval probability of first DMR. Patient-specific fitted trajectories are obtained from

$$\hat{\mu}_i(t) = \beta_0 + \beta_1 \log(1+t) + \beta_2 \{\log(1+t)\}^2 + b_{0i} + b_{1i} \log(1+t).$$

These fitted trajectories can be plotted against observed log-MRD values and used to estimate individualized DMR probabilities over future monitoring intervals.

1.9 Benchmark Models

Because a Stan interface is not currently installed in the workspace, the pipeline also fits benchmark models using installed R packages. The interval-censored Weibull model uses

$$T_i \sim \text{Weibull}(\lambda_i, \kappa), \quad \log \lambda_i = \delta_0 + \delta_1 y_{i0}$$

for the molecular-only model, with a complete-case extension

$$\log \lambda_i = \delta_0 + \delta_1 y_{i0} + \delta_2 \text{age}_i + \delta_3 \text{male}_i + \delta_4 \text{duration}_i.$$

The longitudinal benchmark model is a mixed-effects model,

$$y_{ij} = f(t_{ij}) + \beta_3 z_{ij} + u_{0i} + u_{1i} t_{ij} + \epsilon_{ij},$$

where $f(t_{ij})$ is a natural spline in years from imatinib start. These benchmark models verify that the cleaned data support interval-censored survival modeling and longitudinal mixed modeling, but the proposed joint model remains the primary model.

1.10 Assumptions and Sensitivity Analyses

The primary assumptions are:

1. the latent log-MRD trajectory is adequately represented by a quadratic function of $\log(1 + t)$ plus patient-specific random effects;
2. reported floor values at $\log\text{-MRD} \leq -5$ are left-censored observations;
3. the DMR hazard is approximately constant within each observed monitoring interval;
4. censoring is conditionally independent given the observed follow-up and modeled molecular trajectory;
5. each patient can experience first DMR only once.

Recommended sensitivity analyses are:

1. fit an exact-value model that treats floor values as observed -5 to assess floor-censoring impact;
2. fit a model without γ_2 to evaluate whether explicit visit-gap adjustment changes inference;
3. replace the quadratic $\log(1 + t)$ trajectory with spline time terms;
4. compare molecular-only and complete-case covariate-adjusted survival models;
5. evaluate prior sensitivity for α , especially using $N(0, 1)$ and $N(-0.5, 1.5^2)$ alternatives.